



FACULTY OF INFORMATION TECHNOLOGY
BACHELOR OF SCIENCE IN INFORMATICS AND COMPUTER SCIENCE
END OF SEMESTER EXAMINATION
ICS 1202: OBJECT-ORIENTED PROGRAMMING I

Date: 11th December 2017

Time: 2 Hours

Instructions

1. This examination consists of **FIVE** questions.
2. Answer **Question ONE (COMPULSORY)** and any other **TWO** questions.

Question One [30 Marks]

- a) Explain the following terms as used in object-oriented programming: **(10 Marks)**
 - i. Class
 - ii. Object
 - iii. Constructor
 - iv. Access modifier
 - v. Interface
- b) Explain the difference between method overloading and method overriding as used in computer programming. **(4 Marks)**
- c) Using code segments, explain how a Java program implements polymorphism in object-oriented programming. **(6 Marks)**
- d) Write a Java class that will implement the class represented in the UML diagram below: **(10 Marks)**

Laptop extends Computer
-weight: float -screenSize: int
+Laptop(name: String, weight: float, screenSize: int) +getWeight(): float +setWeight(weight: float): void +showDetails(): void

Question Two [15 Marks]

- Using code segments, explain the concept of data abstraction as used in object-oriented programming. **(4 Marks)**
- Using code segments, explain the concept of data hiding as used in object-oriented programming. **(4 Marks)**
- Write a simple Java application that will create two objects of class Cylinder and displays their area and perimeter. **(7 Marks)**

Cylinder extends Circle
-height: int
+Cylinder(radius: int, height: int) +getArea(): float +getPerimeter(): float

Question Three [15 Marks]

- a) Explain the significance of polymorphism as used in object-oriented programming. **(2 Marks)**
- b) Explain the difference between polymorphism and inheritance as used in object-oriented programming. **(4 Marks)**

```
public class Book {  
  
    private String title;  
    private int isbn;  
  
    public Book(String title, int isbn) {  
        this.title = title;  
        this.isbn = isbn;  
    }  
  
    private void showDetails() {  
        System.out.println("Title: "+title);  
        System.out.println("isbn: "+isbn);  
    }  
}
```

```
public class Novel {  
  
    private String author;  
  
    public Novel(String title, int isbn) {  
        super(title, isbn);  
        this.author = author;  
    }  
  
    public void showDetails() {  
        super.showDetails();  
        System.out.println("author: "+author);  
    }  
}
```

- c) Identify 3 errors in the code segments below. **(6 Marks)**
- d) Write the correct Java codes for class Book and Novel. **(3 Marks)**

Question Four [15 Marks]

- a) Explain the difference between an abstract class and a concrete class as used in object-oriented programming. **(4 Marks)**
- b) Explain the difference between a final method and a static method as used in object-oriented programming. **(4 Marks)**
- c) Using code segments, illustrate how an abstract class can be used by a concrete class in Java programming. **(7 Marks)**

Question Five [15 Marks]

- a) Explain the main purpose of designing a graphical user interface for a computer program. **(2 Marks)**
- b) Explain at least 3 differences between AWT and Swing components which are Java graphical user interface components. **(6 Marks)**
- c) Using a code segment, explain the syntax of a try-catch-finally close. **(7 Marks)**